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MODELLING THE INFLUENCE OF CELL SHAPE AND OTHER MECHANICAL CUES ON DIFFERENTIATION

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INTRODUCTION

The extent of spreading of a cell confined to a square island depends on the dimensions of the island[1]. Cells confined to smaller islands spread to a limited extent and are more likely to form adipocytes, while cells confined to larger islands spread out more and usually tend to form osteoblasts[2]. In this study, we propose a model that essentially utilizes the differences in the cytoskeletal arrangements in cells confined to adhesive islands of different sizes, to predict the corresponding differentiation trend observed in the experiments, and subsequently extend the model to investigate experimental observations involving the impact of other mechanical 'cues' on cell spreading and differentiation[4,5].

MODEL OVERVIEW: A STATISTICAL MECHANICS APPROACH

Consider a thermodynamic system that consists of the cell and the adhesive square island, interacting with the external, thermo-nutrient reservoir. A given cell shape is characterized by a 'spread micro-state', which is defined as the connection of material points on the cell membrane to those on the island.

At every spread micro-state, the cell is at mechanical and thermal equilibrium with its surroundings. Defining the free-energy of the cell G^i_{tot} in micro-state *i* as

$$G^i_{\text{tot}} = G^i_{\text{cyto}} + G^i_{\text{elas}} + G^i_{\text{wink}} \tag{1}$$

where G^i_{cyto} is the free-energy associated with the cytoskeletal configuration in the cell in micro-state *i* (as defined by Vigliotti et al[3]), G^i_{elas} is the contribution from the passive elasticity of the cell (modified hyperelastic Ogden model) and G^i_{wink} is the free-energy associated with the winkler foundation used to approximate the adhesion between the cell and the island. We assume that the islands are printed on rigid substrates, so that the free-energy associated with the deformation of the substrate is zero, and hence does not feature in eqn. (1).

The external reservoir imposes a homeostatic constraint on the cell such that the cell fluctuates between a variety of different micro-states with their associated shapes and free-energies, while ensuring

$$\sum_i G^i_{\text{tot}} = G_0 \tag{2}$$

where the summation denotes the ensemble average of the total free-energy of the cell, while G_0 denotes the free-energy associated with the homeostasis constrained imposed by the external reservoir.

With the equilibrium free-energy of the cell in a micro-state given by eqn. (1), the probability P^i that the cell is observed in micro-state *i* (with total free-energy G^i_{tot}) can be calculated by minimizing the free-energy of the micro-states with respect to the constraint in eqn. (2) to yield

$$P^i \propto e^{-\beta G^i_{\text{tot}}} \tag{3}$$

where β is the parameter chosen to enforce the homeostatic condition imposed on the system by the thermo-nutrient reservoir.

Since the cell can take a wide variety of shapes on each island, the cell is allowed to explore this enormous phase-space through a few million Markov Chain Monte Carlo (MCMC) steps on each island. A small random displacement is given to the cell in each MCMC walk, for which the total-free energy of the cell is calculated using eqn. (1) and the resulting configuration (micro-state *i+1*) is accepted with the probability $e^{-\beta(G^{i+1}_{\text{tot}} - G^i_{\text{tot}})}$.

RESULTS

Simulations were performed with cells confined to square islands of a range of sizes and the extent of spreading on each island was observed. Fig. 1 depicts a few shapes (or spread micro-states) of cells observed on islands of two different sizes. Notice that the island area (A^p) is normalized with A^e – the area of cell under zero elastic strain.

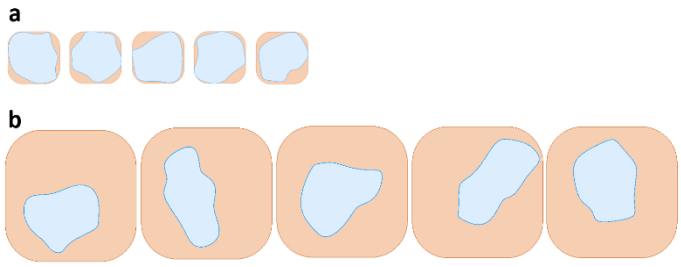

Figure 1: Few of the spread micro-states observed in cells on (a) a small island (A^p/A^e=1), and (b) a large island (A^p/A^e=8)

Our hypothesis is that studying the fluctuations of cells between different cytoskeletal configurations when confined to an island of a given size plays a key role in predicting the trends in cell spreading and differentiation. Fig. 2(a) depicts the G_{cyto} distribution as a function of patch size. In order to correlate the information contained in the cytoskeletal free-energy distribution to cell differentiation, we develop a differentiation model where the central parameter is the mean cytoskeletal free-energy $\bar{G}_{\text{cyto}}$ calculated over a long period of time period for a cell confined to a given island. For a cell fluctuating on an island over N_c spread micro-states,

$$\bar{G}_{\text{cyto}} = \sum_{i=1}^{N_c} P^i G^i_{\text{cyto}} \tag{4}$$

For large N_c, central limit theorem can be applied to eqn. (4) to express $\bar{G}_{\text{cyto}}$ as a normal distribution $N(\mu, \bar{\sigma})$, where $\bar{\sigma} = \frac{\sigma}{\sqrt{N_c}}$, μ and σ are the mean and variance of the cytoskeletal free-energy distribution of the cell on the given island. Fig. 2(b) depicts the distribution of $\bar{G}_{\text{cyto}}$ corresponding to the G_{cyto} distributions illustrated in Fig. 2(a).

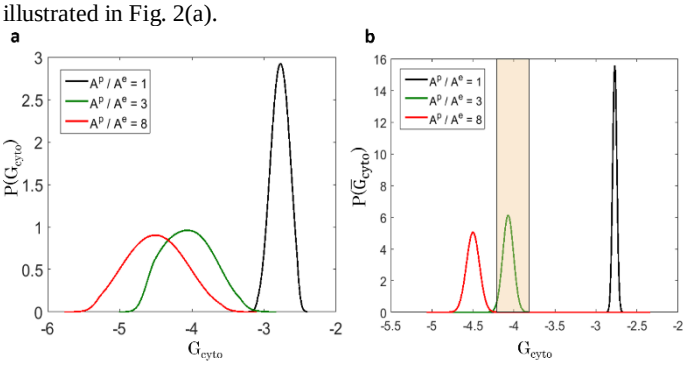

Figure 2: (a) G_{cyto} and (b) $\bar{G}_{\text{cyto}}$ distributions for three different island sizes (x-axis normalized with G_0)

Let us assume that the cells undergo differentiation when their mean cytoskeletal free-energy $\bar{G}_{\text{cyto}}$ calculated over a long period of time period attains a value over a specific range. For example, stem cells differentiate into osteoblasts when $\bar{G}_{\text{cyto}}$ lies within the interval $[G_{\text{osteo}} - \Delta_{\text{osteo}}, G_{\text{osteo}} + \Delta_{\text{osteo}}]$ (this interval is illustrated as the yellow box in Fig. 2(b)). Therefore, the probability that a cell differentiates into an osteoblast can be written as

$$P_{\text{osteo}} = \int_{G_{\text{osteo}}-\Delta_{\text{osteo}}}^{G_{\text{osteo}}+\Delta_{\text{osteo}}} p\left(\bar{G}_{\text{cyto}}\right) d\bar{G}_{\text{cyto}} \tag{5}$$

An expression similar to eqn. (5) can be written for other cell types, with cell-specific differentiation parameters. In this study, we focus on two types of cells formed – osteoblasts and adipocytes. Therefore, G_{osteo}, Δ_{osteo}, G_{adipo}, Δ_{adipo} and N_c are the different parameters that determine the number of cells that commit to the two lineages on each of the adhesive islands. For a particular choice of the five parameters, we plot the differentiation as a function of island size as predicted by the simulation (2D) in Fig. 3(a), and compare it with the experimental observation (3D) in Fig. 3(b).

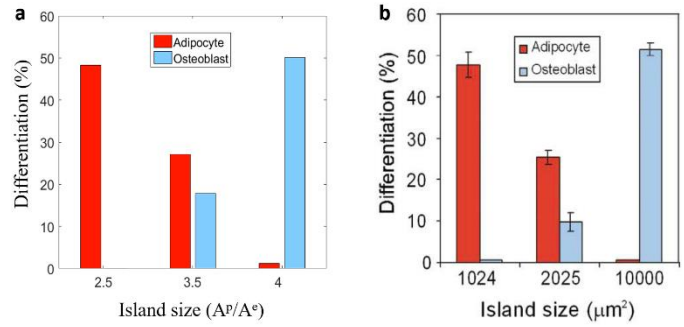

Figure 3: Differentiation as a function of island size, from (a) simulation and, (b) experiment.

DISCUSSION

We observe that the statistical mechanics model, coupled with the differentiation model, is able to capture the essential trend observed in the differentiation of stem cells as a function of the size of the adhesive island. The range of areas of islands over which the differentiation trend is predicted by the simulation is lesser than the range observed in the experiments. This can be attributed to the fact that the spreading of cells is significantly higher in 3D, but the model used in the current study only captures the 2D effects, since the model's primary aim is to demonstrate its versatility while being computationally efficient.

The versatility of the model stems from the fact that it can be used to study different types of mechanical 'cues', such as island size, island shape and substrate stiffness. With the same underlying statistical mechanics model and the different parameters that dictate the differentiation behavior, the current study is extended to predict the differentiation of cells as a function of substrate rigidity (as reported in Engler et al[4]) and the aspect ratio of the adhesive island (as reported in Kilian et al[5]). The contractility of the actomyosin bundles can also be varied to study the impact of actin or myosin inhibitors such as cyto-D or blebbistatin on the extent of differentiation[2,4,6].

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